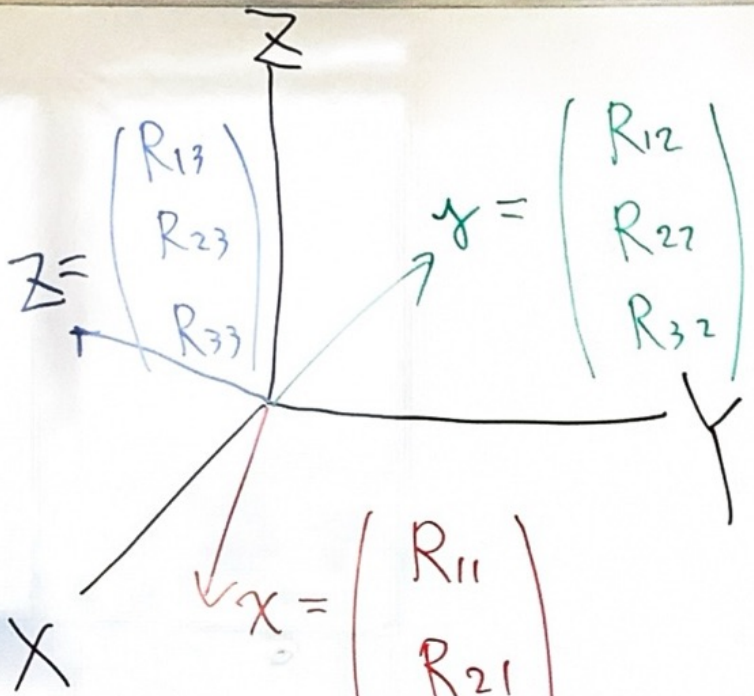


$$Z = \begin{pmatrix} R_{13} \\ R_{23} \\ R_{33} \end{pmatrix} \quad Y = \begin{pmatrix} R_{12} \\ R_{22} \\ R_{32} \end{pmatrix} \quad X = \begin{pmatrix} R_{11} \\ R_{21} \\ R_{31} \end{pmatrix}$$

$$\begin{cases} J\dot{\omega} = J\omega \times \omega + \tau \quad (\tau:\text{トルク}) \\ \dot{R} = R\,\Omega(\omega) \end{cases}$$

$\dot{R}$：姿勢の変化　　R：現在姿勢

ω が変化 → ω を更新

$$\omega = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

$$R_{新} \Leftarrow R_{古} + \dot{R} \times \Delta t \quad (\Delta t = 0.01)$$

（RをΔtごとに更新）

$$R(t) = \begin{pmatrix} R_{11(t)} & R_{12(t)} & R_{13(t)} \\ R_{21(t)} & R_{22(t)} & R_{23(t)} \\ R_{31(t)} & R_{32(t)} & R_{33(t)} \end{pmatrix}$$

回転行列（方向余弦行列）